Multiscale darkness detection for crack area fraction measurement in scanning electron micrographs of Inconel 718

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Abstract

Quantifying crack area in scanning electron microscopy supports the characterisation of defects in metallic microstructures. This study examines a deterministic multiscale darkness detector for measuring the two-dimensional crack area fraction in Inconel 718 micrographs. The evaluated pipeline combines background correction, multiscale candidate detection, morphological cleanup and connected-component filtering without a learned region classifier. The primary collection comprises 31 whole fields. At the reported MAD multiplier of 6.0, the predicted crack area fraction was 1.827%, compared with a reference value of 1.803%. The predicted-to-reference area ratio was 1.0133, corresponding to a net difference of 0.024 percentage points and an aggregate relative discrepancy of approximately 1.3%. The reported background specificity was 98.86%. Among seven displayed classical method summaries, the darkness detector had the area ratio closest to unity under the original selection procedure. A field-level comparison reported a smaller absolute area error at the selected setting than at the default setting in 23 of the 31 fields.

This revision adds two things the previous version asserted without showing. First, the operating-setting sweep is now reported in full: the earlier grid stopped at the default setting with the area ratio still falling, and extending it identifies 6.0 as the point at which the reported and reference areas agree. Second, the area endpoint is placed against the range it can resolve, using a reconstruction of the reference annotation that is exactly correct about position and wrong only about width. The revision also adds a cross-modality evaluation: the same detector, unchanged, applied to 31 transmission X-ray microscopy frames of austenitic steel, where it reports 2.473 times the annotated crack area pooled across four specimen groups and clears an all-background prediction on one of them.

These findings support the detector as an interpretable approach to aggregate crack area measurement within the evaluated collection. The principal result concerns agreement in total image area, with pixel-level segmentation considered as a separate endpoint.

Keywords: scanning electron microscopy; Inconel 718; crack area fraction; multiscale image analysis; deterministic segmentation; cross-modality transfer

1 Introduction

Quantitative microscopy links microstructural observations to material condition and processing history. Crack area is one useful descriptor because it summarises the fraction of an observed field occupied by annotated cracking. Extracting this quantity from scanning electron microscopy (SEM) images requires a consistent definition of the foreground and a documented relationship between image processing and the final measurement. Recent work on microstructure inference and microscopy-specific representation learning demonstrates the growing role of automated image analysis in materials characterisation [1,2].

A deterministic detector offers a directly inspectable route from image intensity to a binary mask. This is attractive when the study concerns a defined material and imaging regime and when investigators wish to examine the intermediate processing stages.

Background correction can separate local contrast from slower variations in illumination, while multiscale processing can accommodate structures with different apparent widths. Morphological and component-based operations then convert the candidate response into the mask used for area measurement.

The evaluation endpoint must match the scientific quantity of interest. Image-analysis validation frameworks distinguish task-specific measurements from general pixel classification scores [3,4]. Area estimation likewise has its own accounting requirements [5]. In the present study, crack area fraction refers to the proportion of pixels in a two-dimensional SEM field. It is distinct from a direct measurement of crack volume or a stereological volume-fraction estimate [6].

Related work. Reporting a segmentation in the units of the downstream measurement rather than as an overlap score is established practice outside materials science: in land-cover accuracy assessment, Olofsson et al. set out the error matrix in units of area proportion together with a bias-adjusted area estimator [5]. Within materials microscopy the same instinct appears in several recent forms. Work on domain-informed segmentation targets argues that classical methods retain an advantage because domain knowledge such as a known phase volume fraction can be imposed explicitly, and evaluates volume-fraction alignment rather than overlap [14]. Segment-anything-based models for material micrographs have added boundary-sensitive scores for the same reason [15]. A 2026 analysis of architecture choice in materials segmentation warns that segmentation errors propagate undetected into porosity fractions and grain-size distributions [16].

Classical, threshold-based crack-area measurement in SEM has direct precedent: Vidal et al. binarise SEM images of chromium electrodeposits with a histogram threshold chosen by a Laplacian-of-Gaussian criterion and report the cracked area [17], and unsupervised segmentation has recently been used for quantitative fractography of quasi-cleavage fractures [21]. Deterministic geometric quantification of two-dimensional SEM fracture morphology across metallic materials has also been reported [20], and classical measurement methodology for SEM has been applied to microscale fatigue crack propagation through moire-based distortion correction [22]. Crack quantification in a different material system has independently reported modest overlap scores on narrow cracks, attributed them to one-to-two-pixel boundary placement, and reported area fraction, width and tortuosity as the primary quantities [18]; the present study reaches the same position from a different direction. A counterpoint worth stating is that non-learned segmentation has been reported to struggle on rough fracture surfaces precisely because a manual intensity threshold varies across features [19], which is the difficulty the background correction and multiscale stages of Section 2.2 exist to address.

The present study claims no novelty for the framing; its contribution is the measurement, made in those units, for one deterministic detector on one alloy, together with the ways the conventional scores mislead about it.

We examine the reported behaviour of a multiscale darkness detector on Inconel 718 whole-field images. The analysis centres on aggregate area agreement at a selected operating setting, the corresponding background specificity, and a descriptive comparison with classical image-processing methods. Representative microscopy panels and a processing diagram connect the numerical results to the evaluated workflow. Two sections added in this revision place the selected setting against the full sweep and against the range the area endpoint can resolve, and a third carries the detector to a second imaging modality.

2 Materials and methods

2.1 Image collection and evaluation units

The primary evaluation comprises 31 whole-field SEM images of Inconel 718 from the collection identified in the source study [7]. The images are 1024 x 1024 pixels. Reference masks define the annotated crack pixels, and the mean reference crack area fraction for the

whole fields is 1.803%. Each field serves as an image-level evaluation unit.

The collection also includes 26 released crops originating from 18 parent images represented among the whole fields. The present primary endpoint uses the whole-field collection. The crop set provides related views of the material and its image framing, and is reported separately wherever it is used.

A provenance limitation of this collection, disclosed here because it affects how the reference should be read. The archive's own class list declares fifteen target classes, of which the only crack class is a ceramic environmental barrier coating. The Inconel 718 folders used here, including the crack class on which this entire evaluation rests, are present in the archive but absent from that class list; the undeclared folders and the Inconel 718 folders are the same set. The masks exist, are pixel-level, and are released under CC BY, but no annotation protocol, annotator count or class definition accompanies them. The collection also publishes no pixel size: there is no resolution tag in the image files, no databar in the fields, and no magnification in the record, and the collection's subsets originate from different microscopes. No scale bar can therefore be drawn on any panel from this collection, and none is shown. The crack area fraction reported throughout is a dimensionless ratio of pixel counts and does not depend on a pixel size; physical lengths cannot be derived from these images and none are reported.

2.2 Detector configuration

The evaluated detector processes an 8-bit grayscale image and applies background correction before identifying dark candidate regions over seven spatial scales. The scales are 5, 10, 20, 40, 80, 160 and 320 pixels. Candidate selection combines a median absolute deviation (MAD) criterion with an absolute intensity gate.

Mask processing consists of hole filling, morphological opening and connected-component filtering, with an opening size of 13 pixels and a minimum retained region area of 40 pixels. Components use 8-connectivity. The learned region classifier is disabled in the scored branch, so the configuration produces its candidate mask without fitting a learned region-classification model. Table 1 lists the reported settings.

Component	Reported setting
Input representation	8-bit grayscale SEM image
Multiscale processing	5, 10, 20, 40, 80, 160 and 320 px
Selected MAD multiplier	6.0
Absolute intensity gate	10.0
Morphological opening size	13 px
Minimum connected-region area	40 pixels
Connectivity	8-connected components
Learned region classifier	Disabled in the evaluated branch

On the number of free parameters. One constant, the MAD multiplier, is exposed and swept, and it is the only one given a grid in any comparison reported here. It is not the only constant. The absolute intensity gate is a second threshold, held fixed rather than fitted in this study. Behind those sit the seven-rung scale ladder and the two size floors, also fixed. A reader comparing this against a single-parameter classical filter should know that the filter carries one constant and this configuration carries a frozen preprocessing budget.

Figure 1 shows the microscopy example and the intermediate image outputs. Figure 2 summarises the scored branch from grayscale input to area measurement.

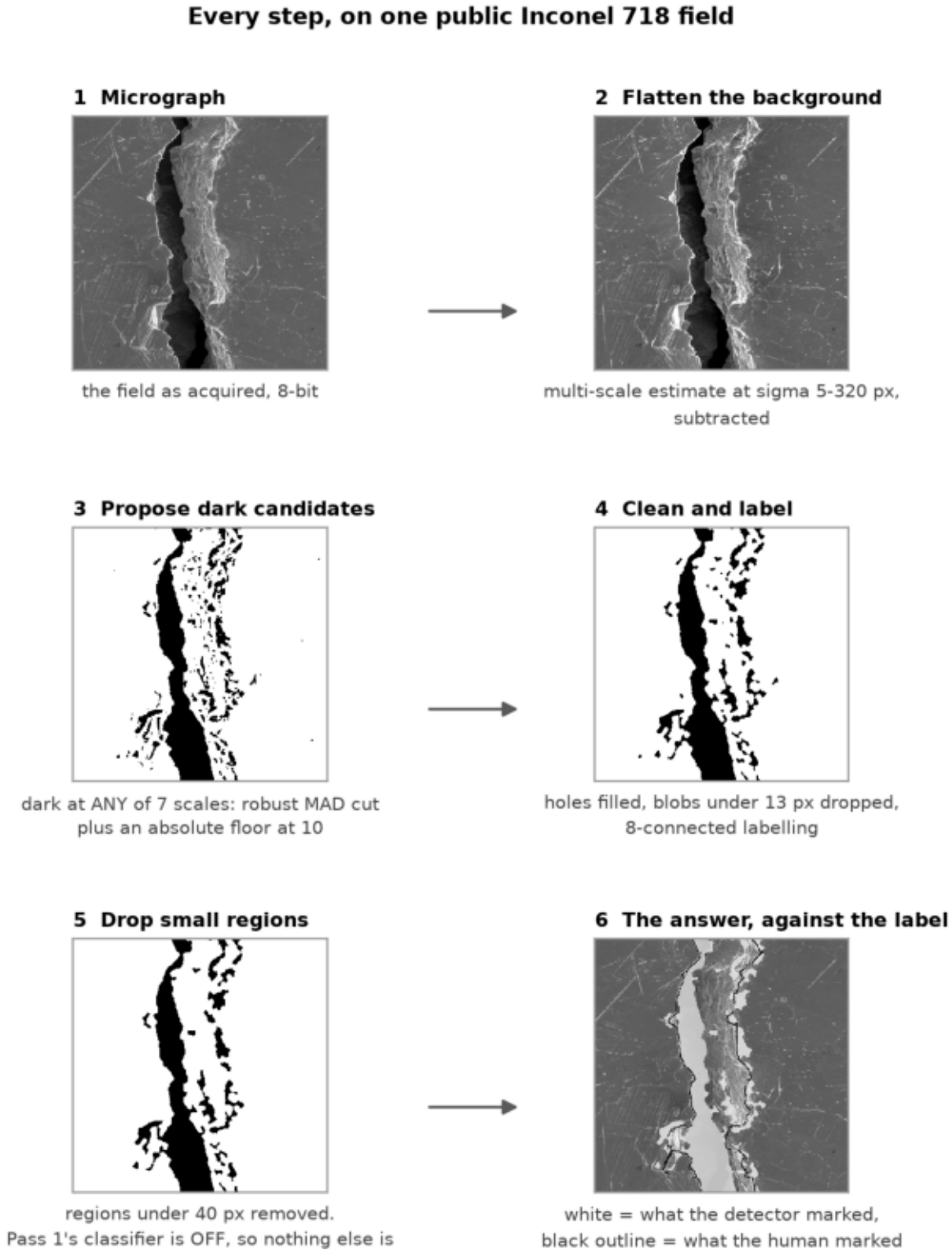


Figure 1. Representative detector outputs for one Inconel 718 SEM field. Panels show the input, background correction, candidate detection and mask processing, and the prediction against the reference annotation. No scale bar is shown because the source collection publishes no pixel size; see Section 2.1.



Figure 2. The scored detector workflow. Candidate detection and mask processing produce the predicted crack area fraction. The learned region classifier is disabled. Subsequent region growth and manual correction are outside the evaluated branch.

2.3 Measurement endpoints

Let N denote the number of evaluated image pixels, Nref the reference foreground count and Npred the predicted foreground count. Reference and predicted crack area fractions, expressed as percentages of image area, are Ax = 100 Nx / N for x in {ref, pred}.

The predicted-to-reference area ratio is R = Apred / Aref and the net area difference is dA = Apred - Aref. R = 1 denotes equality of the reported foreground area totals. dA is expressed in percentage points. The absolute relative aggregate area discrepancy used for the comparator summary is Earea = 100 |R - 1|, expressed as a percentage of the reference area. It is computed from the displayed aggregate ratio and is distinct from the mean of individual relative image errors.

Background specificity describes the fraction of reference background pixels classified as background. With TN and FP denoting true-negative and false-positive pixels, specificity is TN / (TN + FP). Aggregate area agreement and background specificity have separate interpretations from foreground overlap [3,4].

An all-background reference point accompanies every accuracy figure in this manuscript. On an image that is 1.803% crack, a prediction that marks nothing scores 98.20% pixel accuracy, so marking a pixel improves accuracy only where more than half of what is marked is correct. Accuracy quoted without that reference point is not interpretable at this prevalence, and both numbers are given together throughout.

2.4 Comparator and field-level analyses

The displayed comparison includes the multiscale darkness detector, a darkness-or-Sato union, Sato filtering, Frangi filtering, Meijering filtering, Sauvola thresholding and percentile thresholding [8-12]. Seven method summaries appear in the source table. Each arm receives its own parameter by the same leave-one-image-out rule, on the same images, through the same scoring code.

The selected operating setting is a MAD multiplier of 6.0. A paired field-level comparison contrasts its absolute area error with that of the default setting. The comparator summaries and field counts are descriptive.

2.5 Cross-modality evaluation

To test whether the configuration is specific to this imaging modality, the identical detector, with the identical constant, was applied to 31 transmission X-ray microscopy (TXM) frames of austenitic stainless steel spanning four specimen groups. Those frames carry hand-made correction masks that mark crack and not-crack explicitly and leave the remainder unreviewed, so only reviewed pixels are scored: 38.1% of all pixels in those frames were ruled on. The reviewed area holds 8,518,853 crack pixels against 96,610,708 not-crack (11.3:1), so a false positive is priced by the annotation rather than assumed. Percentages in Section 4 are therefore percentages of the reviewed area, not of the frame.

TXM frames carry a pixel size of 29.24 nm/px, which is derived from mosaic geometry — a 30 micrometre viewing window at 0.35 tile overlap over a 9 x 5 mosaic — and not read from instrument metadata. Scale bars on TXM panels use that value and scale with it if the acquisition geometry is corrected.

3 Results

3.1 Aggregate crack area agreement

At a MAD multiplier of 6.0, the detector produced a reported crack area fraction of 1.827%, compared with a reference fraction of 1.803%. The resulting area ratio was 1.0133. The net difference was +0.024 percentage points, corresponding to an aggregate relative excess of approximately 1.3%.

The selected setting also recorded a reported background specificity of 98.86%. This value characterises rejection of non-crack background pixels. Together, the area ratio and specificity quantify two aspects of the output: the total marked area and the classification of the background population. Figure 3 places the result against every collection scored in this work, and Table 2 collects the principal numerical results.

Endpoint	Reported or derived value
Reference crack area	1.803% of image area
Predicted crack area	1.827% of image area
Predicted-to-reference area ratio	1.0133
Net aggregate area difference	+0.024 percentage points
Absolute relative aggregate area discrepancy	Approximately 1.3%
Background specificity	98.86%
Pixel accuracy	97.79%
Pixel accuracy of an all-background prediction	98.20%
Fields with lower absolute area error than the default	23 of 31

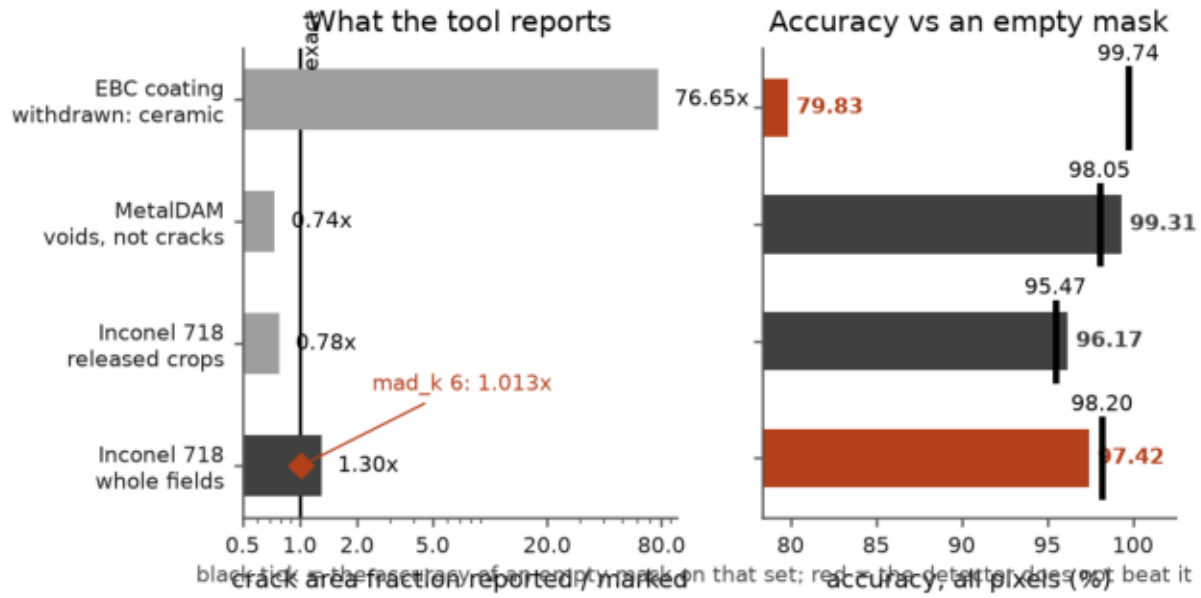


Figure 3. Reported crack area fraction against the annotated fraction, and pixel accuracy against the accuracy of an all-background prediction, for every collection scored in this work. A ratio of 1.0 is exact agreement; the diamond marks the selected setting on the whole fields. MetalDAM is included for reference and labels voids rather than cracks.

3.2 Descriptive comparison of aggregate area estimates

The darkness detector had the reported area ratio closest to unity among the seven displayed method summaries. Its ratio of 1.013 corresponds to an absolute relative aggregate area discrepancy of approximately 1.3%. The darkness-or-Sato union had a ratio of 1.029. The closest classical arm on its own was Sato ridge at 0.885. Table 3 and Figure 4 retain the complete displayed comparison.

arm	marked%	area rep./true	accuracy	specificity
this detector	1.83%	1.013x	97.79%	98.86%
darkness OR Sato	1.86%	1.029x	97.77%	98.84%
Sato ridge	1.60%	0.885x	97.98%	99.07%
Frangi ridge	2.04%	1.132x	97.71%	98.71%
Meijering ridge	1.54%	0.852x	98.00%	99.12%
Sauvola local	1.36%	0.754x	98.16%	99.29%
percentile threshold	0.88%	0.488x	98.18%	99.54%

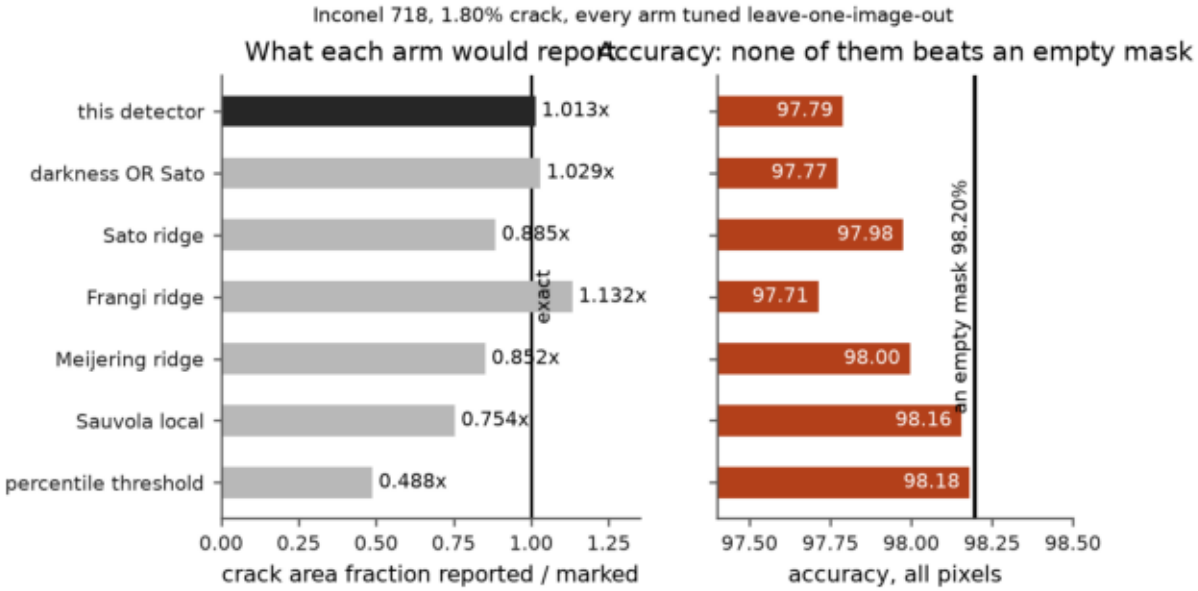


Figure 4. The eight compared arms on the whole fields, ranked by how close the reported crack area fraction is to the annotated one, with pixel accuracy and the all-background reference point beside it. The detector places first on area agreement. No arm clears the all-background accuracy on this collection; three do on the released crops.

The area comparison identifies a specific strength of the selected configuration: its total foreground area closely matches the reported reference total. The numerical ordering applies to that endpoint and to these displayed summaries. It does not establish superiority in foreground overlap, and Section 5.1 states what it does not establish.

3.3 Field-level area error comparison

A lower absolute area error was recorded at the selected setting than at the default setting in 23 of the 31 whole fields, approximately 74% of the evaluated fields. The count supplements the aggregate comparison by describing the direction of the paired image-level difference.

3.4 The operating-setting sweep, in full

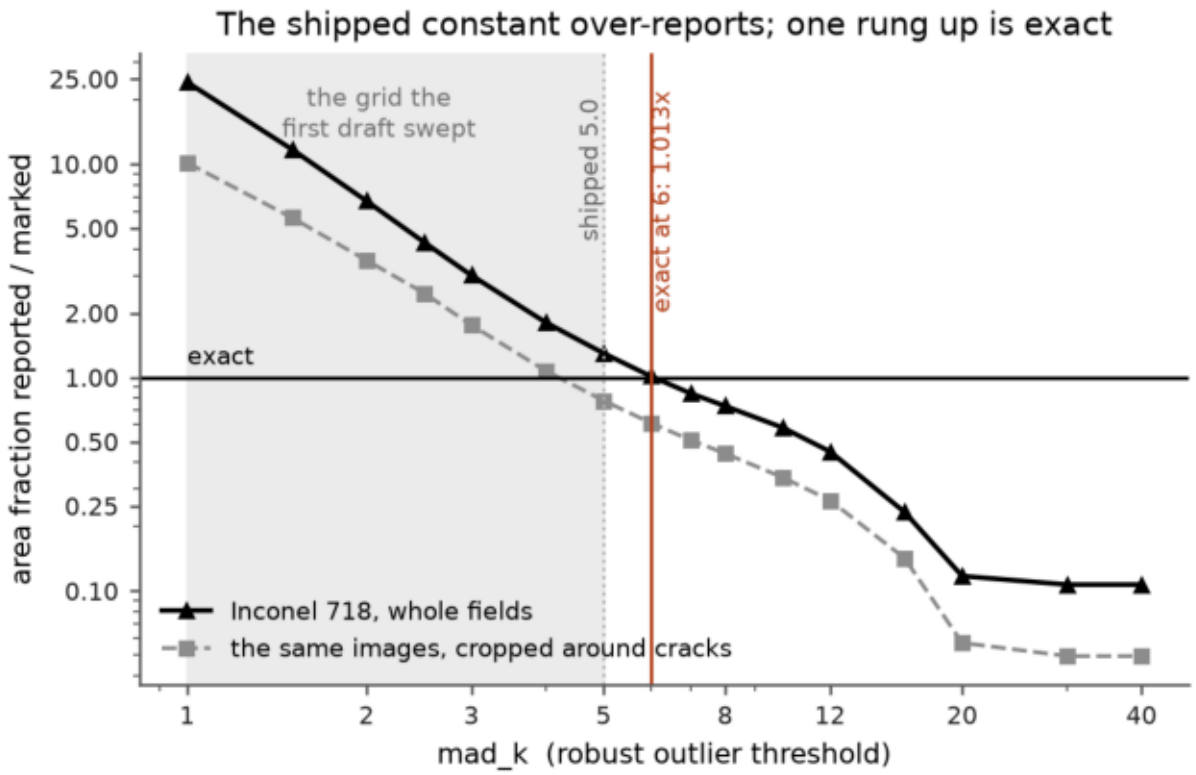


Figure 5. Reported crack area fraction divided by the annotated fraction, against the detector's MAD multiplier, on the whole fields and on the released crops of those same fields. A value of 1.0 is exact agreement. The shaded band marks the grid swept in the earlier analysis, which stopped at the default setting. Both axes are logarithmic.

The earlier analysis selected its operating setting from a grid of seven values running from 1.0 to the default of 5.0. The default was the largest value tested, and the area ratio was still falling when the grid ran out. A selection made from a grid whose maximum is the value being selected cannot fail, and this one did not.

Extending the sweep gives the two-sided curve. The reported area ratio falls monotonically with the multiplier, crossing unity between 5.0 and 7.0, and the crossing is at 6.0, where the reported fraction is 1.827% against the annotated 1.803%. At the earlier default the detector over-reports by 1.298 times; one rung higher it agrees to 1.3%; one rung beyond that it under-reports at 0.842 times. Across the 31 whole fields the absolute area error is smaller at 6.0 than at the default in 23 of 31 images, paired Wilcoxon $p = 0.0124$.

Two consequences should be read with the result. The direction of the useful change is towards marking less, not more. And on the released crops the same change moves the area ratio the wrong way, from 0.93 to 0.72 ($p < 0.0001$), because those crops are cut around their own cracks and the detector already under-reports on them. The two framings of the same images disagree about the correct operating setting, which is a reason not to read a single tuned constant as a property of the method.

3.5 What the area endpoint can resolve

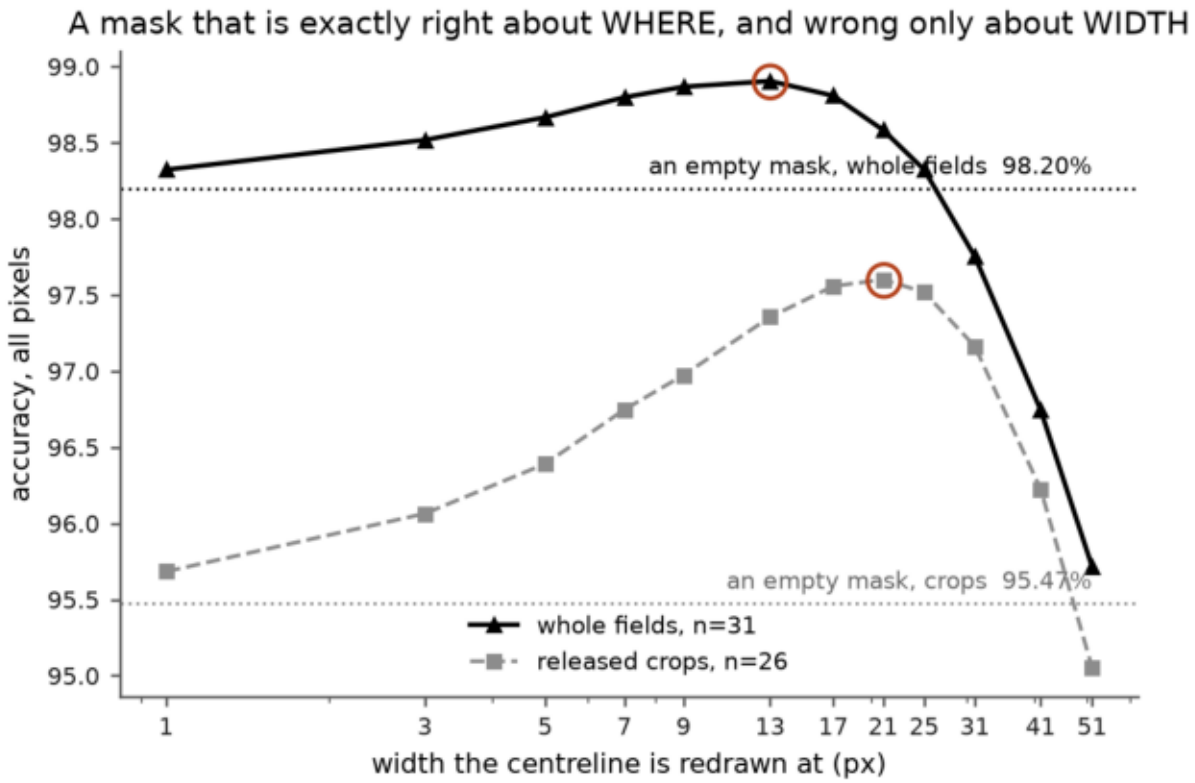


Figure 6. Pixel accuracy of the reference annotation reduced to its centreline and redrawn at a fixed width: a prediction that is exactly correct about where every crack is and wrong only about how wide. Dotted lines give the accuracy of an all-background prediction on each collection. Circles mark each curve's maximum.

A comparison is only readable against the range in which it can move. Taking the reference annotation itself, reducing it to its centreline and redrawing it at a fixed width gives a prediction with perfect localisation whose only error is width. Whatever that construction scores is an upper reference point for this collection.

On the whole fields the entire range is 0.708 percentage points. The reconstruction peaks at 98.905% pixel accuracy at a drawn width of 13 px, against 98.197% for an all-background prediction. A prediction with perfect localisation and the best available width is worth less than one point of accuracy over marking nothing. That is not a statement about any detector; it is a statement about what pixel accuracy can express at this prevalence, and it is why the area endpoint rather than accuracy carries the result.

On the released crops, where the annotated fraction is 4.53%, the range is 2.128 points — a reconstruction ceiling of 97.601% against 95.473% — and there the comparison becomes informative. The detector reaches 96.70%, which is 57.6% of the available range.

The same construction locates the width. The reconstruction reproduces the annotated crack area fraction at 13 px on the whole fields and 21 px on the crops, so a mask drawn near the annotation's own stroke width already carries approximately the correct amount of foreground, and the residual disagreement concerns placement rather than quantity.

4 Cross-modality evaluation

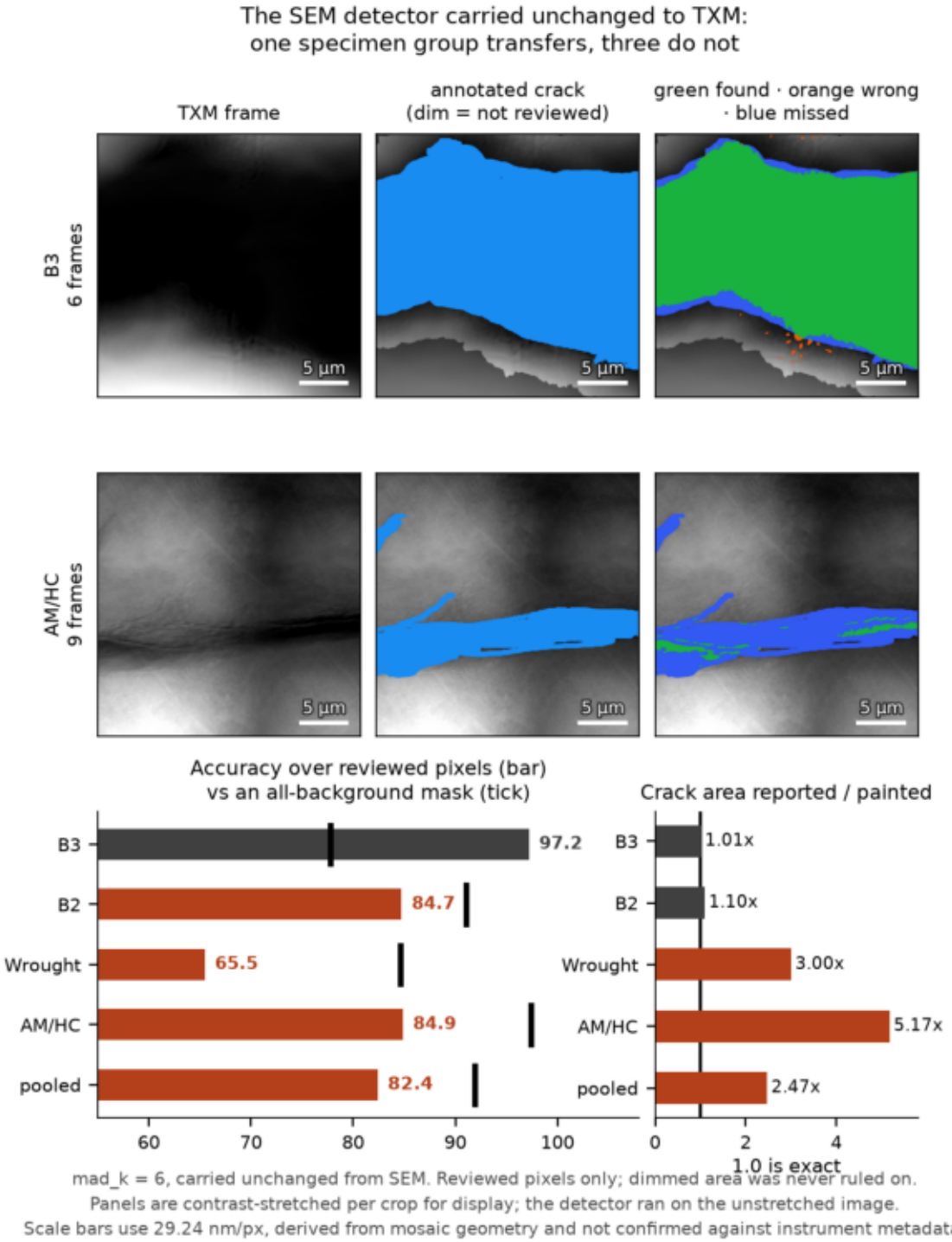


Figure 7. The same detector, with the same constant, applied to transmission X-ray microscopy of austenitic steel. Panels: one frame from each of the two specimen groups whose behaviour differs most, showing the frame, the annotated crack, and the prediction decomposed into found, incorrectly marked and missed. Dimmed area was not reviewed and is excluded from every number. Bars: pixel accuracy over reviewed pixels against the accuracy of an all-background prediction, and the reported crack area fraction against the annotated one, for all four specimen groups and pooled. Scale bars 5 micrometres, from a pixel size derived from mosaic geometry; see Section 2.5.

The detector was carried unchanged to 31 TXM frames. Over the reviewed area, crack is 8.103% and an all-background prediction scores 91.897% accuracy.

Pooled, the configuration does not transfer. It records 82.378% accuracy, -9.52 percentage points below an all-background prediction, and reports 2.473 times the

annotated crack area at a recall of 0.649. It clears both no-information comparators comfortably — an Otsu darkness cut with no model records 23.1% and marking every pixel 8.1% — so the configuration is doing something. It is not doing enough to improve on marking nothing.

The pooled figure is four different results. The specimen groups do not behave alike:

Specimen group	Frames	Crack % of reviewed area	Accuracy	All-background	Recall	Area reported / annotated
B3	6	22.2%	97.2%	77.8%	0.942	1.01
B2	12	8.9%	84.7%	91.1%	0.195	1.10
Wrought	4	15.4%	65.5%	84.6%	0.878	3.00
AM/HC	9	2.6%	84.9%	97.4%	0.138	5.17

One group transfers. On B3 the detector records 97.2% accuracy against 77.8% for an all-background prediction, a margin of 19.4 points, at a recall of 0.942 and an area ratio of 1.01. On B2 the reported area is close to correct at 1.10 while the recall is 0.195, so the right quantity of foreground is placed substantially in the wrong locations. On AM/HC the detector reaches a recall of 0.138 while marking 13% of the reviewed area, and reports 5.17 times the annotated crack area. On Wrought it finds most of the annotated crack and over-reports the area by 3.00 times.

This is reported as a transfer result and not as a capability. Four specimen groups give four answers, each group is a small number of frames, and the groups are different materials and loading conditions, so the pooled figure is a weighted mean over four distinct measurements rather than an estimate of one. The acquisition parameters for this collection — instrument, beam energy, voxel size at the specimen, reconstruction algorithm — are not recorded with the data, and the pixel size used for the scale bars is derived rather than measured. Any physical length derived from these frames inherits that assumption. What the section establishes is narrow and still worth stating: a constant fitted on SEM steel in one laboratory, carried without modification to a different imaging physics, clears both no-information comparators on all four groups and clears an all-background prediction on one.

5 Discussion and conclusions

5.1 Interpretation of the principal measurement

The principal finding is close aggregate area agreement for a deterministic detector applied to the evaluated Inconel 718 collection. A reference fraction of 1.803% and a predicted fraction of 1.827% differ by 0.024 percentage points. Expressing the result in both image-area units and relative reference-area units makes the measurement scale explicit. The ratio of 1.0133 supplies an interpretable summary for readers comparing marked area with the reference total.

The complete displayed comparator set places this result in context. Among the seven summaries, the darkness detector has the smallest absolute deviation of the area ratio from unity. The field-level count provides a complementary observation: the selected setting has lower reported absolute area error than the default in most of the evaluated whole fields. Section 3.4 adds the information that the earlier selection could not have identified any other setting, because its grid stopped at the default; the extended sweep locates the crossing and the selected setting survives it.

5.2 Interpretation of the processing workflow

The evaluated branch uses image intensity, multiple spatial scales and explicit mask operations. Each stage has a visible intermediate representation in Figure 1, and Figure 2 separates the operations that lead to the final area estimate. Disabling the learned region classifier gives the reported configuration an inspectable processing path without a learned

region-classification step.

The reported specificity of 98.86% indicates high background rejection in the reference background population. This observation is consistent with the intended use of a darkness criterion to identify a relatively small foreground population. It complements the area measurement while retaining its own definition and denominator.

5.3 Conclusions

A multiscale darkness detector achieved close reported agreement with the aggregate reference crack area in 31 Inconel 718 SEM whole fields. At a MAD multiplier of 6.0, the predicted area was 1.827% against a reference of 1.803%, giving a ratio of 1.0133 and an approximately 1.3% relative aggregate discrepancy. The detector also recorded 98.86% background specificity and the closest area ratio to unity among the seven displayed comparator summaries.

Two results reported here qualify that conclusion rather than supporting it. The pixel accuracy at the selected setting, 97.79%, is below the 98.20% of an all-background prediction, and Section 3.5 shows that the entire range available to pixel accuracy on this collection is 0.708 percentage points, so no configuration could have been distinguished on that endpoint. And the same detector carried to a second imaging modality clears an all-background prediction on one specimen group of four.

Together these establish an aggregate area result and an interpretable workflow within the evaluated collection, and they bound how far either should be read.

6 Limitations and future improvements

6.1 Spatial interpretation and measurement scope

Agreement in aggregate area can arise when false-positive and false-negative areas cancel. It does not establish accurate crack localisation or foreground overlap. At the selected setting the reported pixel accuracy is 97.79%, against approximately 98.20% for an all-background prediction. A decomposition of the aggregate summaries gives approximately 1.56% false-positive and 1.02% false-negative image area. Their near cancellation explains the small net area discrepancy. The 98.86% specificity must not be interpreted as foreground segmentation accuracy. Complete per-image precision, recall, Dice, intersection over union and boundary measures are required to assess localisation [3,4], and are deliberately not reported here because they cannot express the measurement this study is about.

The ranking reported in Section 3.2 is specific to the area endpoint and can be obtained without detection. Dilating this detector's own output, which reads no new evidence from the image, moves it to the top of a conventional overlap ranking while increasing the reported crack area fraction to 2.54 times the reference. That observation is the reason this manuscript reports area agreement and background rejection rather than overlap scores, and a reader should apply the same scepticism to the area ranking: it is a ranking of total marked area and nothing more.

Diagnostic observations further limit the interpretation of the darkness criterion. Approximately 42% of annotated crack area lies outside the reported proposal support. The unreached labelled regions have higher reported mean intensity than the detected dark cores — a mean grey level of 127 against a background of 82 and detected cores of 59 — which is consistent with a contrast-polarity limitation rather than a threshold setting. Panels of the most heavily cracked fields indicate that these regions are the illuminated interior surface of an open crack. Expansion experiments increased marked area and did not establish a validated correction. A future detector could combine dark-core evidence with independently validated contextual or polarity-aware features; such changes require new experiments and joint assessment of area bias and spatial overlap.

6.2 Scale, units and the absence of a scale bar

No panel drawn from the Inconel 718 collection carries a scale bar, because the collection publishes no pixel size in any form: no resolution tag in the image files, no databar in the fields, no magnification in the record, and subsets originating from different microscopes. This is stated rather than left as an omission. The reported quantity is a dimensionless ratio of pixel counts and does not depend on a pixel size, so the principal result survives the absence; no physical length is reported for this collection and none can be.

The TXM panels in Section 4 do carry scale bars, from a pixel size of 29.24 nm/px that is derived from mosaic geometry rather than read from instrument metadata. Those bars, and any physical length derived from those frames, scale with the acquisition geometry if it is corrected.

6.3 Validation and reproducibility

The reference annotation for this collection has undocumented provenance. As recorded in Section 2.1, the Inconel 718 crack class is present in the archive but absent from the collection's own class list, and no annotation protocol, annotator count or class definition accompanies it. No inter-annotator agreement can be computed, and no second annotator exists, so the reference cannot be distinguished from one annotator's judgement. A new benchmark should fix the tuning objective within training folds, preserve complete per-image results, report uncertainty for the primary endpoint, and document the annotation rules.

The released crops share parent images with the whole fields, so they do not form an independent test population. Image framing also changes the reported area ratio: cropping these images around their own cracks moves it from 1.30 to 0.78 with no change to the method, and reverses which operating setting is preferable.

The implementation requires a reproducible specification of the MAD normalisation, the logical combination of the intensity gates, image and mask resizing, the morphological footprint and the software version. The reported 13-pixel opening does not establish whether that value denotes a radius, diameter or another footprint convention.

The current conclusion is limited to Inconel 718 whole fields and a two-dimensional area endpoint, with the cross-modality result of Section 4 scoped as stated there. Generic defect labels from another material collection, including MetalDAM [13], are not interchangeable with crack annotations. Independent material-specific validation is needed before interpreting image area as a physical three-dimensional defect quantity.

Data and code availability

The source study identifies the dataset record in Ref. [7]. The TXM frames of Section 4 are not part of a public release, and the acquisition parameters required to interpret them physically are not recorded with the data. No archival DOI or commit hash is quoted for the analysis code in this revision; both are required before submission, and until they exist a reader cannot independently reproduce the numbers reported here.

References

[1] Durmaz AR, Muller M, Lei B, Thomas A, Britz D, Holm EA, Eberl C, Mucklich F, Gumbsch P. A deep learning approach for complex microstructure inference. Nature Communications. 2021;12:6272. doi:10.1038/s41467-021-26565-5.

[2] Stuckner J, Harder B, Smith TM. Microstructure segmentation with deep learning encoders pre-trained on a large microscopy dataset. npj Computational Materials. 2022;8:200. doi:10.1038/s41524-022-00878-5.

[3] Maier-Hein L, Reinke A, Godau P, et al. Metrics reloaded: recommendations for image analysis validation. Nature Methods. 2024;21:195-212. doi:10.1038/s41592-023-02151-z.

[4] Reinke A, Tizabi MD, Baumgartner M, et al. Understanding metric-related pitfalls in image analysis validation. Nature Methods. 2024;21:182-194. doi:10.1038/s41592-023-02150-0.

[5] Olofsson P, Foody GM, Herold M, Stehman SV, Woodcock CE, Wulder MA. Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment. 2014;148:42-57. doi:10.1016/j.rse.2014.02.015.

[6] ASTM International. ASTM E562-19e1 Standard Test Method for Determining Volume Fraction by Systematic Manual Point Count. West Conshohocken, PA: ASTM International; 2020. doi:10.1520/E0562-19E01.

[7] Tian J. Cross-Material Few-Shot Segmentation for Generalizable Microstructure Analysis. Version 1. Zenodo; 2026. Dataset. doi:10.5281/zenodo.23045328.

[8] Sauvola J, Pietikainen M. Adaptive document image binarization. Pattern Recognition. 2000;33:225-236. doi:10.1016/S0031-3203(99)00055-2.

[9] Sato Y, Nakajima S, Shiraga N, et al. Three-dimensional multi-scale line filter for segmentation and visualization of curvilinear structures in medical images. Medical Image Analysis. 1998;2:143-168. doi:10.1016/S1361-8415(98)80009-1.

[10] Frangi AF, Niessen WJ, Vincken KL, Viergever MA. Multiscale vessel enhancement filtering. In: Medical Image Computing and Computer-Assisted Intervention MICCAI 1998. Lecture Notes in Computer Science. 1998;1496:130-137. doi:10.1007/BFb0056195.

[11] Meijering E, Jacob M, Sarria JCF, Steiner P, Hirling H, Unser M. Design and validation of a tool for neurite tracing and analysis in fluorescence microscopy images. Cytometry Part A. 2004;58A:167-176. doi:10.1002/cyto.a.20022.

[12] van der Walt S, Schonberger JL, Nunez-Iglesias J, et al. scikit-image: image processing in Python. PeerJ. 2014;2:e453. doi:10.7717/peerj.453.

[13] Luengo J, Moreno R, Sevillano I, Charte D, Pelaez-Vegas A, Fernandez-Moreno M, Mesejo P, Herrera F. A tutorial on the segmentation of metallographic images: Taxonomy, new MetalDAM dataset, deep learning-based ensemble model, experimental analysis and challenges. Information Fusion. 2022;78:232-253. doi:10.1016/j.inffus.2021.09.018.

[14] Prakash A, et al. ExpertSegmentation: Segmentation for microscopy with domain-informed targets via custom loss. Acta Materialia. 2025. doi:10.1016/j.actamat.2025.120993.

[15] Tu Y, et al. SAMM: a general-purpose segmentation model for material micrographs based on the segment anything model 2. Advanced Powder Materials. 2026. doi:10.1016/j.apmate.2026.100404.

[16] Lu M, Tripathi PK, Shteyn M, Ganguly D, et al. Context Determines Optimal Architecture in Materials Segmentation. arXiv:2602.04154. 2026.

[17] Vidal C, et al. Analysis of SEM digital images to quantify crack network pattern area in chromium electrodeposits. Surface and Coatings Technology. 2016. doi:10.1016/j.surfcoat.2015.11.049.

[18] Tegetmeyer-Kleine T, Schmitt T, Aquino P, Rahe C, et al. Data-efficient crack quantification in lithium-ion cathodes using foundation model transfer. arXiv:2608.27162. 2026.

[19] Ngo A, et al. Image-based fracture surface defect characterization methods for additively manufactured Ti-6Al-4V tested in fatigue. JOM. 2024. doi:10.1007/s11837-024-06655-7.

[20] Jiao Y, et al. Curvature-driven quantification of 2D SEM fracture morphology across metallic materials. Materials Characterization. 2026. doi:10.1016/j.matchar.2026.116319.

[21] Zheng Z, et al. Quantitative fractography of quasi-cleavage fractures via unsupervised image segmentation. Engineering Failure Analysis. 2026. doi:10.1016/j.engfailanal.2026.111235.

[22] Chen L, Zhang Q, Xue L, Li Y, Xie H, Zhang Q. Distortion correction method for SEM

images based on sampling moire patterns and its application to characterization of the microscale fatigue crack propagation. Measurement Science and Technology. 2025. doi:10.1088/1361-6501/adbc08.

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